

Prospects for Verifying Relational Quantum Mechanics at Macroscopic Scale: Computational Challenges and the Simulation Barrier

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Abstract

Relational Quantum Mechanics (RQM) proposes that the quantum state is not an observer-independent property of a system, but rather a relation between systems. This perspective is conceptually powerful, yet its empirical assessment at macroscopic scales faces an immediate computational obstacle: any faithful simulation of highly entangled many-body dynamics grows exponentially with the number of qubits. This note argues that, for this reason, a full classical verification of relational structure in large-scale quantum systems remains out of reach and is likely to remain so for a long time. At the same time, it outlines a practical workaround — *Staged Classical Pre-Filtering* (SCPF) — that makes restricted investigations feasible on present-day hardware.

1 Introduction

Relational Quantum Mechanics (RQM), introduced by Carlo Rovelli in 1996, advances a strong claim: the state of a quantum system is not an objective physical property in isolation, but a relation between the system and an observer. This shift in viewpoint is philosophically and physically significant. However, it also creates a severe practical difficulty. If one wishes to test the theory in a serious way, one must simulate systems with strong entanglement structure across many subsystems, potentially involving thousands of qubits.

At that point one encounters a familiar barrier: the classical cost of representing and evolving such systems becomes prohibitive long before the physically interesting regime is reached.

2 The Exponential Bottleneck

The core issue is the representation of the quantum state vector. For a system of n qubits, the number of complex amplitudes is exactly 2^n . This scaling is manageable only for small n .

Scenario	Qubits	Memory	Interpretation
Without optimization	36	~ 1 TiB	A raw model with 2^{36} amplitudes in <code>complex128</code> precision already exceeds the capacity of ordinary workstations.
With active SCPF	12	~ 64 KiB	Multi-stage filtering reduces the effective space to 2^{12} states, enabling immediate algorithmic inspection.

This is not a marginal improvement. It is the difference between an intractable simulation and a compact computation that fits comfortably within commodity memory.

3 Staged Classical Pre-Filtering in Practice

Within the QHDA Labs Wildfire Risk PL project, I developed a method that I refer to as *Staged Classical Pre-Filtering* (SCPF). It is not a single optimization, but a layered classical screening procedure applied before each quantum stage. In combination, these layers reduce the computational search space by several orders of magnitude.

3.1 Label Filtering: EFFIS + Heuristic Fallback

Before a grid cell enters the model, it is screened against historical wildfire data obtained through the EFFIS API. Cells with no meaningful risk signal are removed classically, without invoking any quantum simulator. When EFFIS is unavailable, a threshold-based fallback rule is used:

```
heuristic_label(summary) -> int(temp > 25 and rh < 40
and wind > 10 and soil < 0.2)
```

3.2 Dimensionality Reduction Before QSVC

Sixteen input features are reduced to four principal components before the data enters the quantum circuit. This step is crucial: every additional qubit doubles the state-space cost. The feature space is therefore compressed from 2^{16} to 2^4 effective states.

```
PCA(n_components=4) + MinMaxScaler(feature_range=(0,  $\pi$ ))
-> ZZFeatureMap(dim=4)
```

3.3 Candidate Pre-Selection Before QAOA

Sensor-placement optimization via QAOA or SamplingVQE operates only on the eight strongest candidates rather than on the full grid. This reduces the state space from 2^N , where N is the number of grid cells, to $2^8 = 256$ states. Runtime on a standard simulator remains below roughly ten seconds.

MAX_CANDIDATES = 8 -> $2^8 = 256$ states -> <1 KB RAM

SCPF is not, to my knowledge, described in the existing literature in this exact form. It emerged from the practical constraints of the project itself. Each of the three layers can be used independently, while the full pipeline is available as an open-source MIT implementation in the QHDA Labs repository ([GitHub repository](#)).

4 Where Classical Simulation Ends

SCPF is a useful workaround, but it is not a complete answer. A full verification of relational structure in complex quantum systems would require manipulation of approximately 10 000 logical qubits or more. This is where the classical approach reaches its absolute limit.

A direct simulation of 10^4 qubits would require memory on the order of 2^{10000} bytes, an amount so large that it dwarfs any realistic physical resource and vastly exceeds the commonly cited estimate of about 10^{80} atoms in the observable universe. No classical computer can perform such a simulation, regardless of software optimization.

As of 2026, quantum processors remain in a transitional period between the NISQ era and the era of fully error-corrected computation. Achieving stable large-scale behavior for 10 000 *logical* qubits — not physical qubits, but logical qubits protected by error correction — is generally placed somewhere within the coming decade by many industry roadmaps. Google, Microsoft, and others communicate ambitious progress, but without guaranteed timelines.

5 Conclusion

Staged Classical Pre-Filtering is, at present, a practical way to investigate relational quantum structure with available infrastructure. It opens a path, but not yet the full gate.

A complete verification of RQM at macroscopic scale will require native quantum hardware with advanced error correction. Classical emulation of such processes is not merely difficult; it is excluded by the structure of the resource scaling itself.

This should not be read as a reason for pessimism. It is a precise indication of the next technological threshold that theoretical physics and quantum information science must cross if relational quantum theory is to receive the level of empirical scrutiny it deserves.

Reading Time

Approximately 6 minutes.

References

- C. Rovelli, “Relational Quantum Mechanics,” *International Journal of Theoretical Physics*, 35(8), 1637–1678 (1996).
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press (2010).

- K. W. Banasiewicz, *QHDA Labs Wildfire Risk PL — reference SCPF implementation*. QH-DALabs, open source (MIT), 2026. Repository: [GitHub repository](#).